SMART WATER QUALITY MONITORING SYSTEM

B.Tech ECE Mini Project Documentation

Development Platform: Base44 AI

Domain: IoT + AI + Web Application + Water Quality Monitoring

Proposed Controller: ESP32

Live Application: aqua-affable-sense-flow.base44.app

Project Overview

The Smart Water Quality Monitoring System is a web-based IoT monitoring concept designed to collect and display important water-quality parameters. The system is designed around water-quality sensors connected to an ESP32, with readings transmitted to a web dashboard for visualization and AI-assisted interpretation.

This mini project demonstrates how Electronics and Communication Engineering concepts such as sensors, embedded systems, wireless communication, data acquisition, IoT, and AI can be integrated with a modern web application.

1. Objectives

- Monitor important water-quality parameters digitally.
- Provide a simple and responsive web dashboard.
- Display sensor readings and historical trends.
- Identify potentially abnormal readings.
- Provide AI-assisted interpretation of available readings.
- Generate alerts for configurable abnormal conditions.
- Provide a foundation for future real ESP32 and sensor integration.

2. Key Features

- **Dashboard:** Displays overall status, parameter cards, sensor status and latest readings.
- **Live Monitoring:** Provides an interface for sensor readings and real-time monitoring architecture.
- **AI Analysis:** Interprets available readings and categorizes the condition as SAFE, WARNING or UNSAFE according to configured monitoring logic.
- **History:** Stores/visualizes readings for trend analysis.
- **Alerts:** Identifies potentially abnormal pH, turbidity, TDS, temperature or sensor conditions.
- **Sensor Management:** Provides a structure for tracking monitoring devices.
- **Location Management:** Supports monitoring of villages, households, schools, tanks, borewells and public water sources.
- **Reports:** Provides a structure for water-quality reporting.

3. Water Quality Parameters

- **pH:** Indicates acidity or alkalinity of water.
- **TDS:** Estimates the amount of dissolved solids.
- **Turbidity:** Indicates water clarity or suspended particles.
- **Temperature:** Measures water temperature.
- **Conductivity:** Indicates the electrical conductivity of water.

4. Hardware Requirements

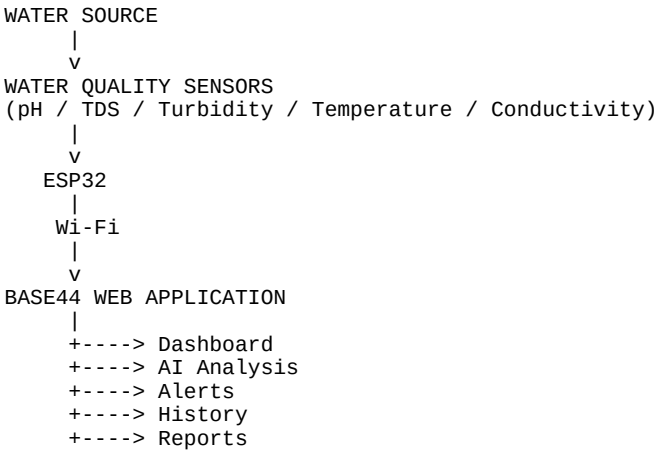
- ESP32 development board
- pH sensor
- TDS sensor
- Turbidity sensor
- Water temperature sensor
- Conductivity sensor

- Power supply
- Water sample/container and appropriate sensor interfaces

5. Software Requirements

- Base44 AI for web application development
- Web browser
- Database/data-storage layer
- REST API or MQTT for possible IoT communication
- ESP32 programming environment for future hardware integration

6. System Architecture



7. Working Principle

- Water-quality sensors collect measurements from a selected water source.
- Sensor outputs are acquired by the ESP32.
- ESP32 can transmit readings through Wi-Fi using an API or MQTT.
- The web application receives and stores the data.
- The dashboard displays current values and trends.
- AI-assisted logic interprets the readings and identifies potentially abnormal conditions.
- Alerts can be generated when configured thresholds are crossed.
- Historical readings can be used for analysis and reporting.

8. Role of Base44 AI

- Base44 AI is used to rapidly build the web application interface and application workflow.
- The platform supports development of dashboard pages, navigation, data views, monitoring interfaces, alert interfaces and responsive UI components.
- The project is designed so that simulated/demo data can be replaced by real ESP32 sensor data in a later version.

9. Applications

- Rural water monitoring
- Schools and colleges
- Water tanks and public water sources
- Household water monitoring
- Agriculture and environmental monitoring
- Educational laboratories
- Community-level IoT monitoring

10. Advantages

- User-friendly web interface
- Supports IoT architecture
- AI-assisted interpretation
- Historical trend monitoring
- Alert capability
- Scalable to multiple monitoring locations
- Strong combination of ECE and software technologies

11. Limitations

- Sensor accuracy depends on sensor quality and calibration.
- Online monitoring can depend on network availability.
- AI interpretation is not a substitute for certified laboratory testing.
- Real deployment requires proper sensor calibration, electrical protection and field validation.
- Water safety decisions should be confirmed with appropriate laboratory testing when required.

12. Future Scope

- Connect actual water-quality sensors to ESP32.
- Implement real-time MQTT communication.
- Add cloud data storage.
- Add mobile notifications.
- Add GPS-based monitoring locations.
- Develop machine-learning anomaly detection.
- Create water-quality maps.
- Support multiple ESP32 monitoring nodes.
- Add predictive analysis using historical data.

- Generate automated monitoring reports.

13. Conclusion

- The Smart Water Quality Monitoring System demonstrates a practical combination of sensors, ESP32, IoT, AI and web technologies. The Base44 AI platform enables rapid development of the web application and dashboard, while the proposed hardware architecture provides a pathway toward real-time physical monitoring.
- The project can be extended into a complete multi-location IoT solution with real sensor acquisition, alerts, historical analysis and intelligent insights.

14. Live Application

The project web application is available at: https://aqua-affable-sense-flow.base44.app

15. GitHub Repository Guidance

Recommended repository structure:

```
Smart-Water-Quality-Monitoring/
■■■ README.md
■■■ documentation/
■   ■■■ project-report.pdf
■   ■■■ system-architecture.png
■   ■■■ screenshots/
■■■ hardware/
■   ■■■ esp32/
■   ■■■ sensor-details/
■■■ diagrams/
```

Keywords: Water Quality Monitoring, IoT, ESP32, AI, Base44 AI, Sensors, TDS, pH, Turbidity, Embedded Systems, ECE, Web Application, Environmental Monitoring